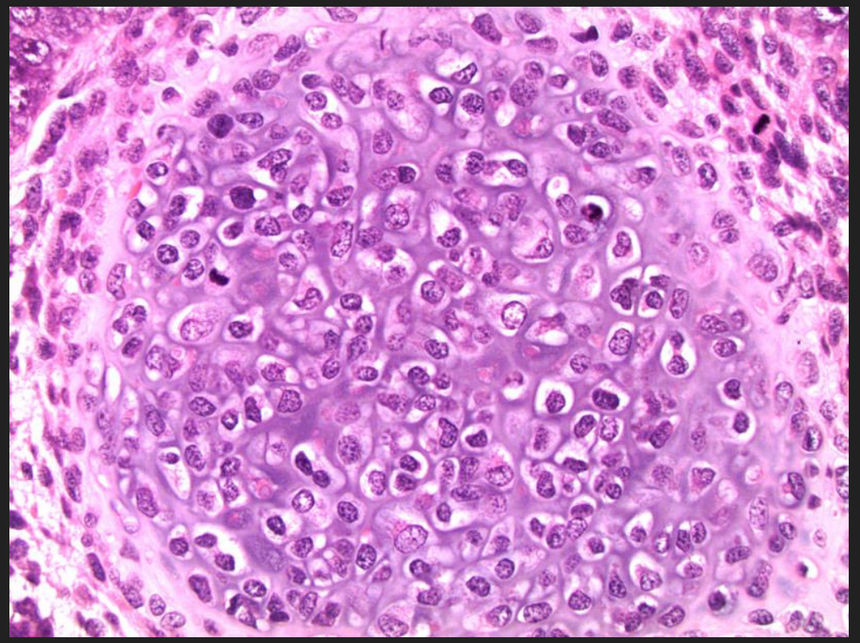
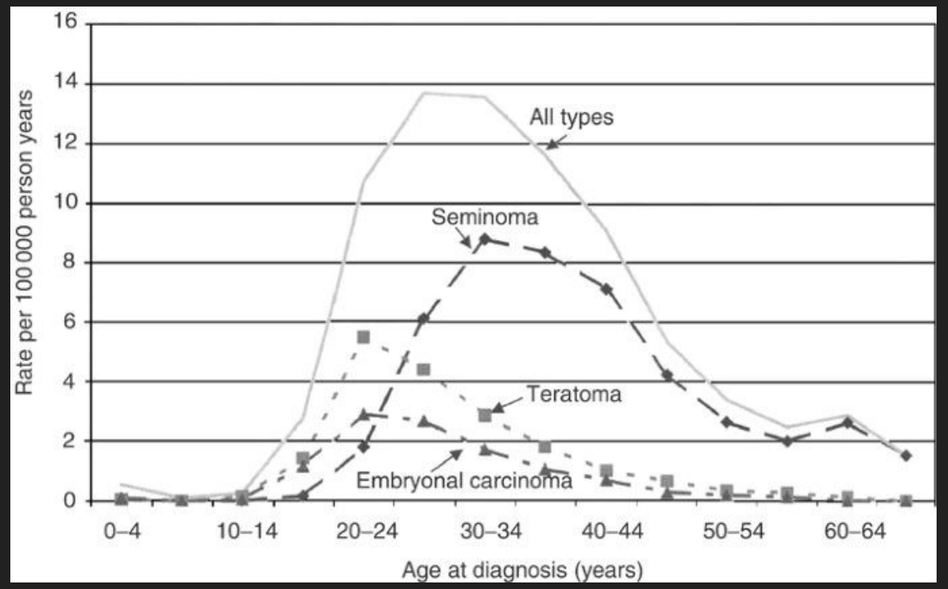
Final Clinical Report



Mir Syed, Emily Moya, Simran Malik

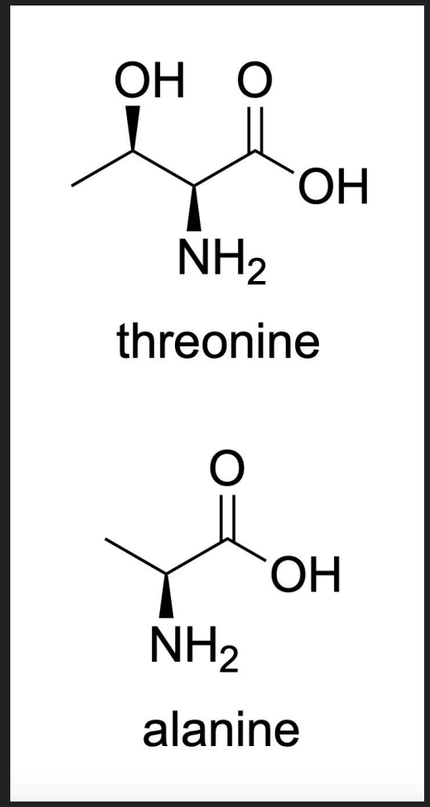
Patient Info & History

- Franc
- 24
- Clearwater, FL
- Male
- Testicular cancer (teratoma in right testicle)
- Stage IIB



Gene Variant #1: AR T878A (androgen receptor)

- Somatic (only detected in tumor tissue)
- Missense variant (T878A: threonine to alanine at 878)
- Gain of function; broadened ligand activation
- Tier III clinical significance in prostate cancer; no documented role in testicular cancer



Gene Variant #2: TERT P308Hfs*43

- TERT (Telomerase reverse transcriptase) functions as a ribonucleoprotein that synthesizes repeats at the end in ends of chromosomes. In cancer cells, TERT mutations prevent telomere shortening which would normally lead to cell cycle arrest or apoptosis. This gives cancer cells the ability to proliferate indefinitely.
- The mutation TERT P308Hfs*43 was somatic since it was only found in tumor cells.
- Deletion of 1 base pair of DNA created a frameshift of the protein at Histidine amino acid position 308. The consequence of this frameshift was truncation of the protein.
- The predicted change of this mutation is inconclusive. In germline mutations of patients with pancreatic cancer, there is evidence that suggest it can lead to loss of Function (PMID: 32923861)
- It’s clinical significance tier is III. The information is inconclusive, yet there is weak/insufficient evidence that suggest it is a possible disease causing mutation. More research would have to be done to form conclusions.

Gene Variant #3: KRAS G12C

- KRAS is a proto-onco gene in the RAS protein family. Essentially, a molecular switch that turns a protein that is on, GTP bound to it, to off, GTP becoming GDP.
- The KRAS mutation was Somatic; it was only found in tumor cells.
- The KRAS gene variant pG12C was a single nucleotide polymorphism.
- The single nucleotide polymorphisms of guanin to tyrosine which case the change of glycine amino acid 12 to cysteine. The protein was not truncated. Glycine and cysteine are both polar and uncharged. The change isn't likely large since they have similar properties.
- The predicted change is a gain of function
- It's clinical significance tier I. KRAS G12C is oncogenic and potentially disease causing.

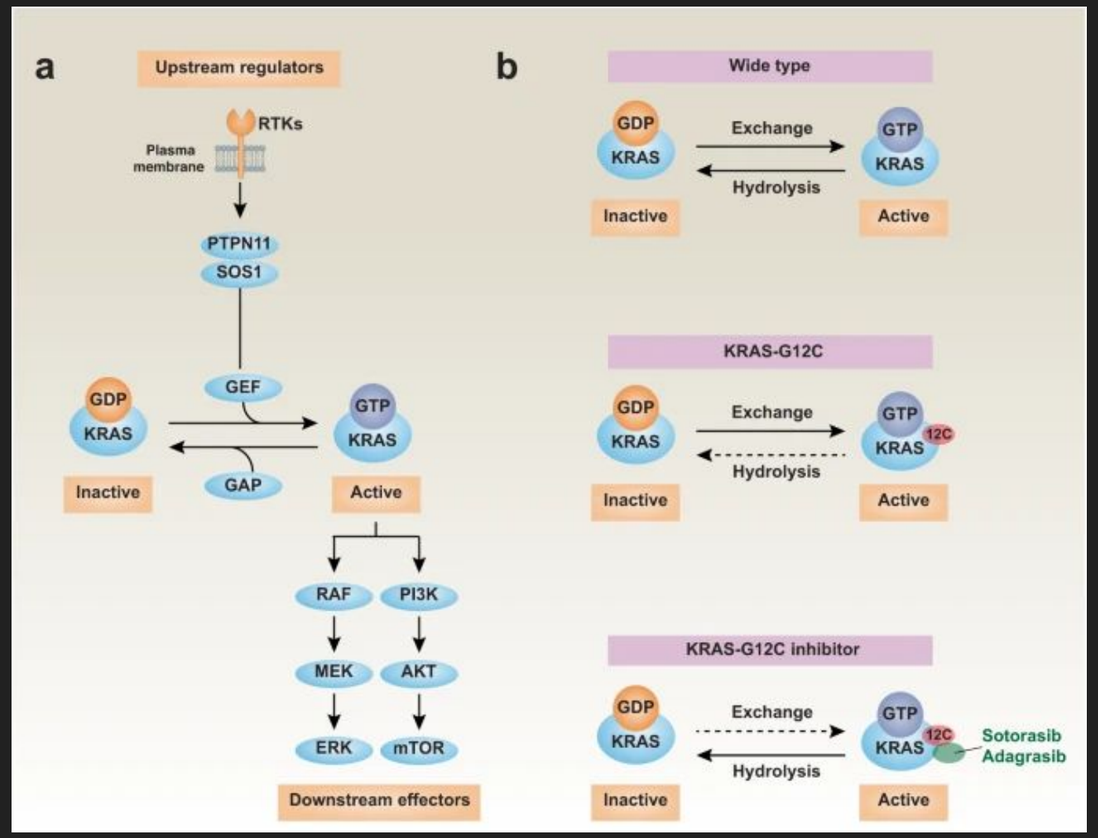


Image of KRAS-G12C mutation which keeps GTP "locked" (Liu et al., 2021)

Clinical Recommendation
Chosen Treatment: Sotorasib (Lumakras)

Gene Variant Association:

This treatment directly targets the KRAS G12C mutation detected in the patient's tumor.

Drug Description and Clinical Use:

Sotorasib is a KRAS G12C inhibitor, its an small cell targeted therapy designed to permanently bind to cysteine at position 12 of the KRAS protein. By pausing the mutated KRAS protein; it stops downstream signaling through the RAS/MAPK pathway,which will stop uncontrolled cell growth and proliferation (PMID: 39047616)

This patient's cancer is not shown in OncoKB due to its rarity. Sotorasib is FDA approved for non-small cell lung cancer with a KRAS G12C mutation and is in advanced clinical trials for colorectal and other tumors that have this same this mutation (both of which hold Tier I clinical significance).

Clinical Recommendation

We selected KRAS G12C inhibitors, (adagrasib or sotorasib) with an potential addition of monoclonal antibody. Since the KRAS mutation identified in the patient is the only actionable driver mutation, identified in the patient is the only actionable driver mutation with tier 1 clinical significance, across multiple cancer types. While this patients cancer is not directly listed on OncoKB; KRAS inhibitors are confirmed in other tumors, which have Tier 1 clinical evidence.

Two Therapeutic options are available

-KRAS inhibitors alone (adagrasib or sotorasib).

-KRAS inhibitor + monoclonal antibody (adagrasib + cetuximab or sotorasib + panitumumab).

Upon research there are no FDA approved therapies for this specific type of tumor , targeting KRAS G12 seems like the only rational targeted therapy

References

- Hemminki, K., & Li, X. (2004, March 16). *Familial risk in testicular cancer as a clue to a heritable and Environmental Aetiology*. Nature News. https://www.nature.com/articles/6601714
- Liu, J., Kang, R., & Tang, D. (2021). The KRAS-G12C inhibitor: activity and resistance. Cancer Gene Therapy, 29, 1–4. https://doi.org/10.1038/s41417-021-00383-9
- Oya, Y., Imaizumi, K., & Mitsudomi, T. (2024). The next-generation KRAS inhibitors…What comes after sotorasib and adagrasib?. Lung cancer (Amsterdam, Netherlands), 194, 107886. https://doi.org/10.1016/j.lungcan.2024.107886
- Walsh, M. F., Sacca, R., Wildman, T., Amoroso, K., Kennedy, J., Zhang, L., Birsoy, O., Mandelker, D., Steinsnyder, Z., Latham, A., Carlo, M. I., Cadoo, K., Kemel, Y., Robson, M., Stadler, Z. K., & Offit, K. (2019). Pathogenic Loss-of-Function Germline TERT Mutations in Patients With Solid Tumors. JCO precision oncology, 3, PO.19.00230. https://doi.org/10.1200/PO.19.00230